

04.01 Radical Expressions

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
A radical contains a perfect-square factor.	Factor the radicand and take out each perfect-square factor.
Terms have matching simplified radicals.	Combine their coefficients like like terms.
A product or quotient contains radicals.	Apply the product or quotient property, then simplify.

Example 1: Simplify a radical

Problem

$$\sqrt{72}$$

How to solve it

Factor:

$$\sqrt{72} = \sqrt{36 \cdot 2}$$

Simplify:

$$\sqrt{72} = 6\sqrt{2}$$

Example 2: Add like radicals

Problem

$$3\sqrt{12} + \sqrt{27}$$

How to solve it

Simplify each term:

$$3\sqrt{12} + \sqrt{27} = 6\sqrt{3} + 3\sqrt{3}$$

Combine:

$$= 9\sqrt{3}$$

Example 3: Multiply radicals

Problem

$$\sqrt{8} \cdot \sqrt{18}$$

How to solve it

Use the product property:

$$\sqrt{8} \sqrt{18} = \sqrt{144}$$

Simplify:

$$= 12$$